

Avalanche Insight Hub

● TECHNICAL EVIDENCE

● RESEARCH AGENDA

MEETING OBJECTIVE

- Show the live platform, the administration view, and the MVP V2 evidence-governance model.
- Distinguish hosted product proof from Swiss RAvaFcast research, Himalayan v3 partner intake, SAR shadow evidence, and future promotion gates.
- Close on whether the SASE/DGRE partner handoff and scientist-led validation program should begin next.

WHAT THIS IS

Governed decision support

Batch-first forecast publication, explicit uncertainty, auditable model-status surfaces, and partner evidence intake.

WHAT THIS IS NOT

Autonomy-first launch

No active MTS-LSTM claim, no promoted SAR claim, no Himalayan accuracy claim, no authority-grade warning claim.

DISCUSSION LANE

Live platform

Public route screenshots and hosted /admin access checks from May 8, 2026.

DISCUSSION LANE

Technical evidence

Governance, benchmark, and candidate-model material available through technical artifacts and administration views.

DISCUSSION LANE

Research agenda

Critical-layer validation, SAR qualification, Himalayan holdout evidence, and candidate-model advancement remain part of the next research program.

Avalanche Forecasting Remains Hard

OBSERVATION SCARCITY

Sparse and discontinuous evidence

Storm windows create the worst blind spots exactly when human access becomes most constrained.

SNOW SCIENCE

Weak layers carry memory

New snow totals alone do not capture persistent weak-layer structure or critical-layer failure risk.

STATISTICS

Rare-event imbalance distorts trust

Naive accuracy can look good while still missing the avalanche days that matter operationally.

OPERATIONS

Authority risk is real

Danger messaging, masking, and consequence framing matter as much as the raw score.

What this means for the platform: batch-first delivery, reduced-confidence cues, masked terrain semantics, and governed claims are features of scientific discipline.

Operational pressure that is still live

MANUAL SNOWPACK WORK DOES NOT SCALE

Field pits remain expensive, dangerous, and geographically sparse.

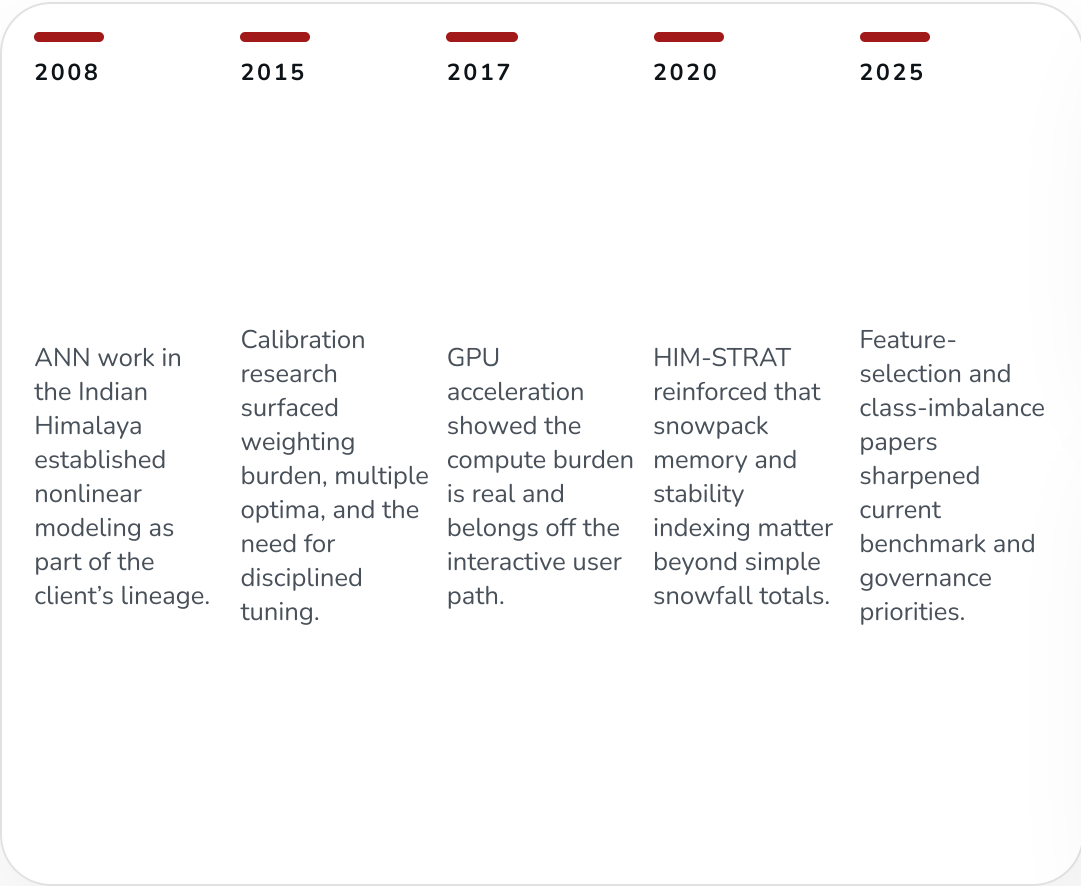
MICRO-CLIMATE VARIABILITY PERSISTS

Regional grids help, but local slope reality still resists broad generalization.

MISSING OCCURRENCE RECORDS WEAKEN VALIDATION

That is why governed event capture and benchmark ownership matter in the next phase.

The Client’s Research Lineage



WHY THIS MATTERS IN THE ROOM

- The scientist team sees continuity with work they already respect.
- The product conversation starts from domain seriousness, not from shiny-tool novelty.
- The current platform is judged as an integration shell around real lineage and real limitations.

WHAT NOT TO CLAIM

No “first-ever” rhetoric

The contribution is in how the stack is packaged, governed, and made discussable now.

WHAT TO CLAIM SAFELY

Continuity with stronger packaging

The platform respects prior research while making evidence surfaces and user-facing semantics easier to inspect.

What Earlier Research Solved, And What It Did Not

ESTABLISHED BY PRIOR RESEARCH

- Feature discipline matters more than feature sprawl.
- Rare-event metrics and class balancing are essential.
- Compute cost is part of the avalanche problem itself.
- Snowpack and weak-layer structure remain first-order science.

IMPLEMENTED IN THE CURRENT PLATFORM

- Published-horizon batch forecast workspace.
- EAWS-style experimental bulletin framing with explicit uncertainty.
- Masked terrain semantics and share or export workflows.
- Governed model-status, provenance, and candidate-gate surfaces.

NEXT RESEARCH PROGRAM

- Scientist-owned critical-layer and weak-layer validation.
- Promotion-worthy candidate-model evidence.
- Operational SAR qualification per region and use case.
- Regional transfer limits and local heterogeneity maps.

What The Live Platform Shows Today

LIVE ROUTE

LIVE PLATFORM /

ADMIN ROUTE /ADMIN

FORECAST SHELL

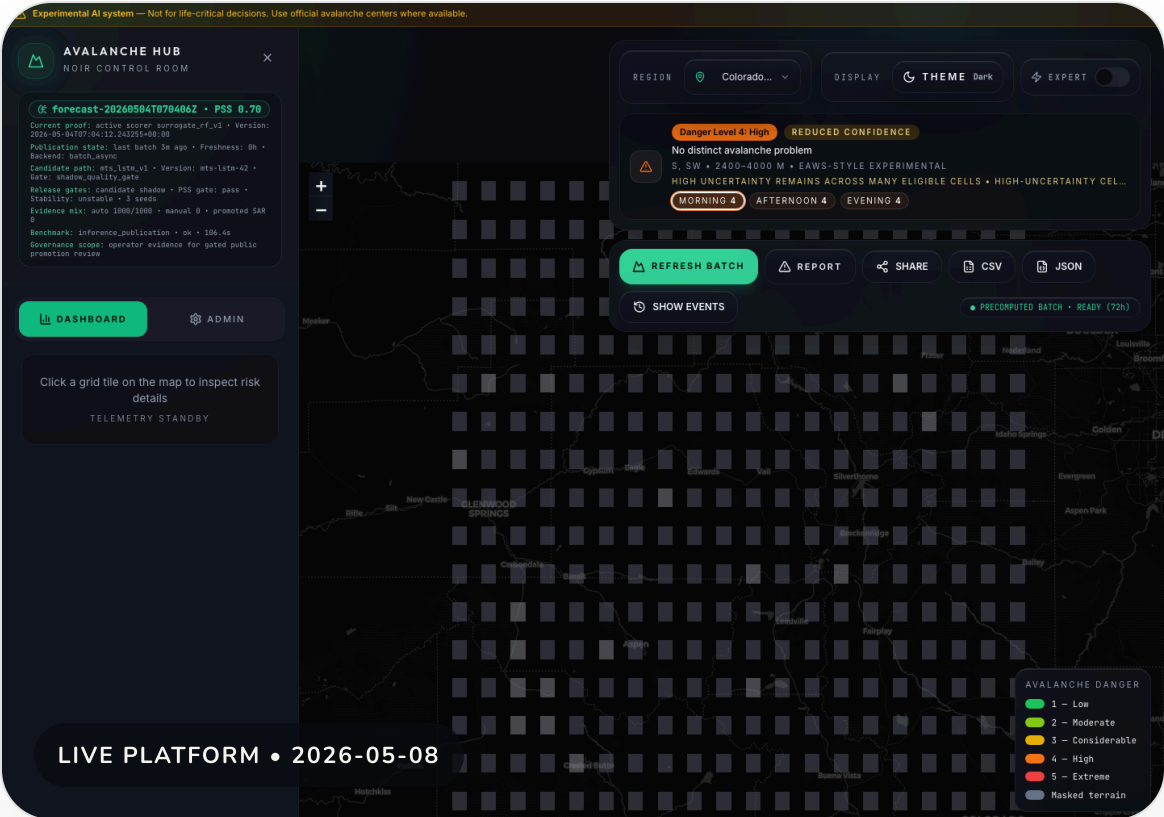
Published artifact workspace

The user sees the latest published batch product rather than waiting on heavy science at click time.

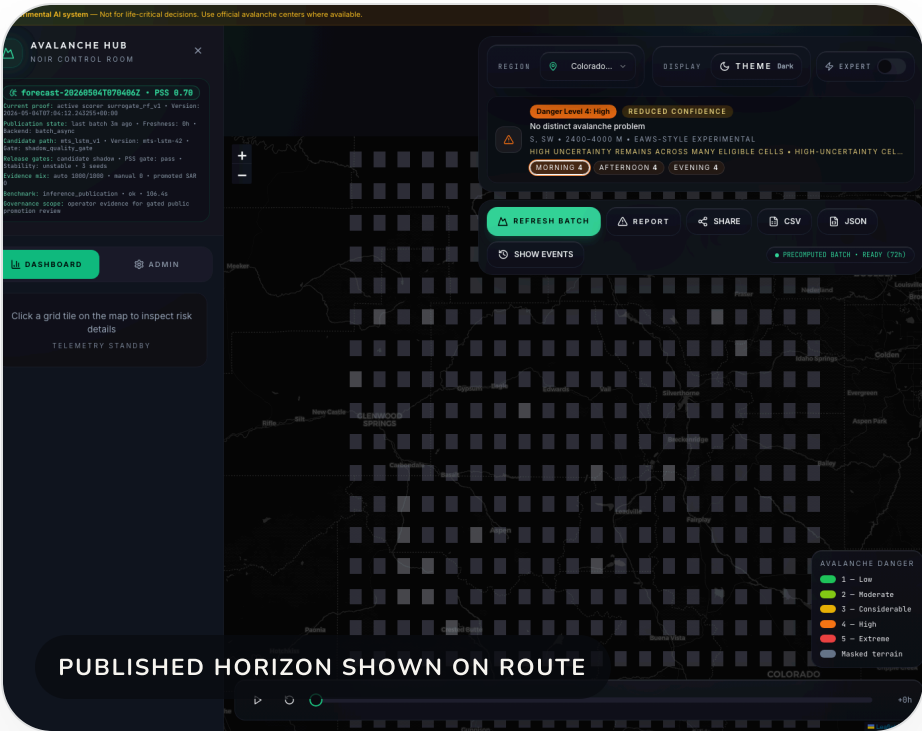
HAZARD COMMUNICATION

Explicit uncertainty and masking

Reduced-confidence language and masked terrain prevent false precision.



Forecast Workspace And Batch- First Delivery



CURRENT BATCH STATE

Current

May 8 proof uses a same-day 20x20 / 72h full-grid technical publication for Colorado Rockies.

SERVING SCORER

RF baseline

Current public route names surrogate_rf_v1 as the active scorer.

WHY BATCH-FIRST MATTERS

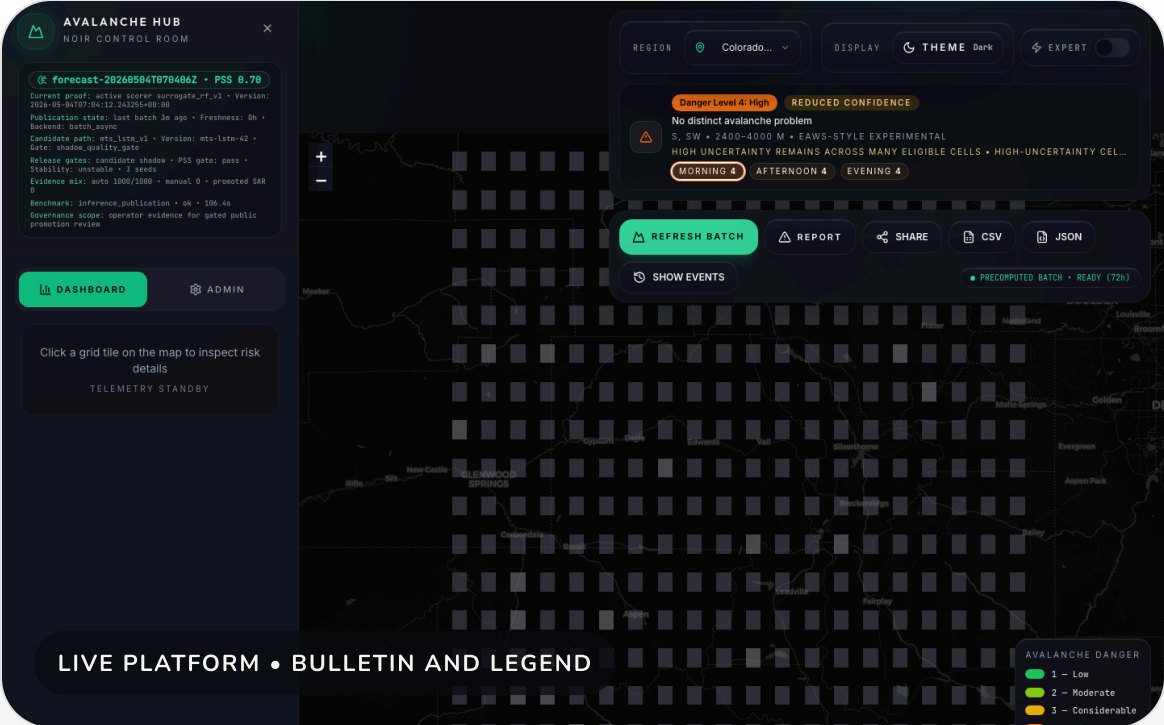
- GPU or heavy candidate workflows stay off the public click path.
- Artifact hydration and lazy hour loading keep the surface responsive while preserving freshness state.

Bulletin Structure, Uncertainty, And Masking

WHAT THE USER CAN READ IMMEDIATELY

- Danger level, avalanche problem framing, daypart, and elevation or aspect cues.
- EAWS-style experimental wording kept visible rather than hidden in notes.
- High-uncertainty and reduced-confidence cues made public instead of buried in admin only.

LIVE PLATFORM



LIVE PLATFORM • BULLETIN AND LEGEND

Share, Export, Report, And Expert Review

STATE HANDOFF

Shareable forecast links

Full-state sharing restores region, time, expert mode, and view state.

ANALYSIS EXPORT

CSV and JSON

Exports let scientists and operators pull forecast evidence out of the UI cleanly.

EVIDENCE INTAKE

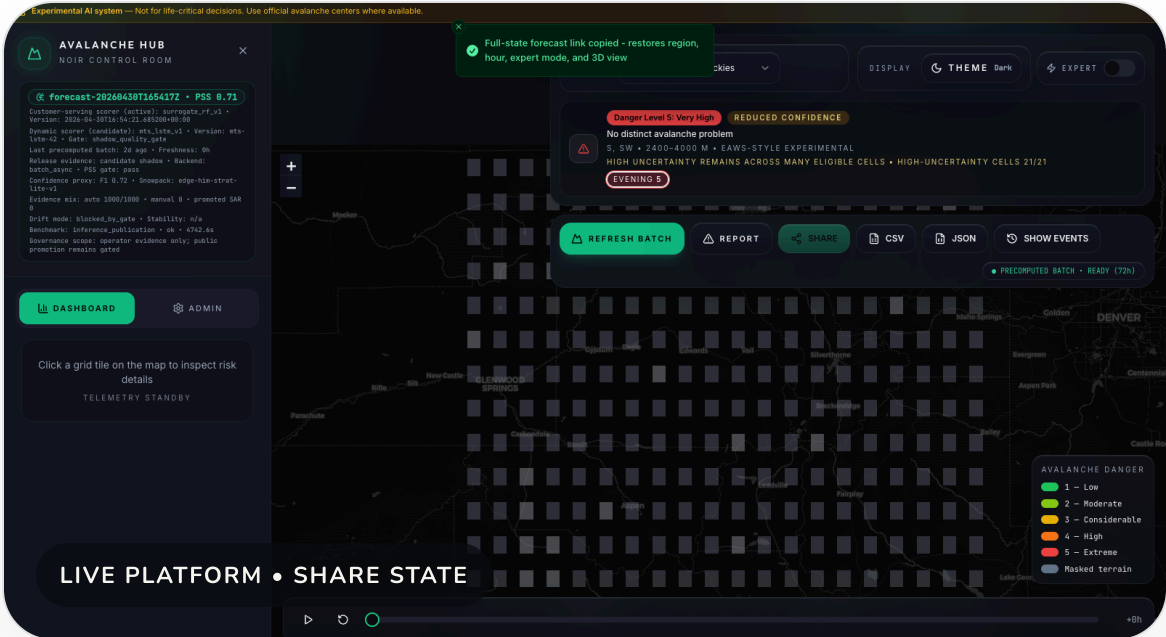
Report flow

Public report capture creates a path toward governed event enrichment rather than one-way display.

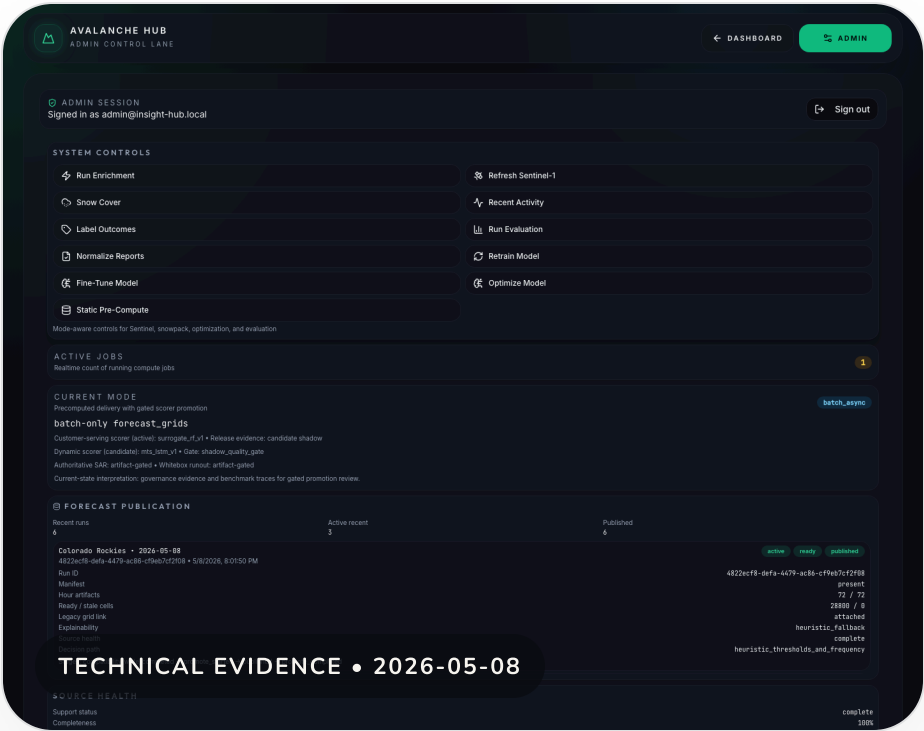
EXPERT REVIEW

Events and overlays

The workflow already anticipates scientist or operator inspection beyond a simple heatmap.



Administration Surfaces And Evidence Context



● LIVE PLATFORM

● TECHNICAL EVIDENCE

Current access and interpretation

Hosted /admin route

Shows administration access and route availability when reachable.

Current access check

Fresh hosted-authenticated access succeeded, so the signed-in observability view can be discussed as technical evidence.

What the administration view adds

Source health, provenance, model status, benchmark traces, stability, and publication state.

What it does not settle

Candidate advancement is not settled here. This is richer observability, not a shortcut around validation.

Fallback rule: if hosted-auth access is unavailable later, revert to the gate screenshot plus reconstructed governance tables.

RF Baseline, RAvaFcast Research, And Modal.com Split

What is active now

Public scorer

surrogate_rf_v1 is active; the current run reports heuristic_fallback, not active TreeSHAP proof.

Candidate lanes

mts_lstm_v1, SAR, and Modal-backed training/segmentation remain gated and off the public scoring path.

Swiss research lane

RAvaFcast is an initial research signal; GPxyz remains blocked until station coordinates exist.

Himalayan evidence lane

The v3 partner contract validates package structure, provenance, and claim locks, not Himalayan accuracy.

Recommended framing: Modal.com supports bounded candidate compute; partner intake and synthetic checks do not authorize GPU, production scoring, or public claim changes.

Governed Evidence Fusion, Not Autonomous Truth Generation

CURRENT GOVERNED PATH

INGEST

Field reports, news, and SAR paths enter with explicit source identity.

GOVERN

`label_confidence`, weights, corroboration, decay, and audit-only rules shape trust.

SUMMARIZE

Admin surfaces persist evidence mix, candidate gates, runtime traces, and stability context.

BLOCK BY DEFAULT

Candidate promotion stays blocked until release artifacts justify stronger claims.

What the current artifact shows and what remains open

Shows

1000 / 1000 positives are autonomous today; 0 are manual positives.

Does not establish

Any human-reviewed truth closure or field-certified autonomous event stream.

Shows

The governed summary spans 6 regions; newest positive source is 2023-12-24 and only `gee_sar` is positive in the sampled artifact.

Does not establish

Fresh, diversified capture or robust regional transfer strong enough for autonomy-forward language.

Benchmark, Stability, And Release Gates

LATEST BENCHMARK

inference_publication

Status is **ok**; latest visible evidence trace is 4742.6s.

STABILITY
CLASSIFICATION

unstable

Current packaged stability evidence is explicitly conservative, not celebratory.

PROMOTION RULE

Blocked
until
earned

Candidate models and Himalayan claims stay gated until local evidence, holdout metrics, release artifacts, and scientist review justify stronger claims.

Current bounded evidence

Seed count

3 seeds only. Read as bounded governance evidence.

Threshold drift

0.157726 is visible and should remain visible in the scientist discussion.

Source of truth

Current admin surfaces, benchmark pack v0, validation protocol v0, v3 partner evidence contract, and the stability summary artifact.

Current reading

“We have explicit benchmark and stability traces, and they are conservative enough to keep stronger claims blocked.”

Avoid saying: “the shadow path is stable and ready for production.” The artifact supports only cautious governance language.

What Is Genuinely Unique Now

OPERATIONAL UX

Published-horizon workspace

A usable forecast shell exists today, not just a model notebook or a static bulletin mockup.

SEMANTIC DISCIPLINE

Masked terrain and reduced confidence

The UI refuses to translate missing support into cosmetic low risk.

INSPECTABILITY

Explainability and provenance in context

Reasoning and lineage are inspectable at the point of use, not hidden in backend reports.

GOVERNANCE POSTURE

Blocked until evidence is earned

The platform keeps candidate activation and evidence lanes explicit instead of flattening them into one story.

What Is Not Unique, And Must Not Be Overclaimed

Do not say

Algorithm novelty

ANNs, SHAP, SAR segmentation, LSTM-style modeling, and GPU workers are not unique by themselves.

Operational status

No active public MTS-LSTM claim, no promoted SAR claim, no Himalayan accuracy claim, and no authority-grade warning status.

Snow science closure

No solved weak-layer or critical-layer validation story.

Autonomy closure

No field-certified autonomous truth-generation claim and no D_tidy-equivalent Himalayan truth yet.

SAFE SUBSTITUTE LANGUAGE

- “Explainable baseline plus governed candidate pathways.”
- “Modal.com supports candidate compute, not the current public platform surface.”
- “SAR is promising but still qualification-bound.”
- “The SASE/DGRE packet is a partner-evidence handoff, not prediction success.”
- “The current value is disciplined integration under sparse-data constraints.”

Current Gaps, Blocked Claims, And The Real Question

BLOCKED CLAIM

No promoted
next-gen
scorer

MTS-LSTM remains
candidate-only until
gates and benchmark
deltas are earned.

BLOCKED CLAIM

No promoted
SAR

Coverage signaling
exists; scientist-grade
qualification still does
not.

BLOCKED CLAIM

No Himalayan
or weak-layer
closure

V3 intake validates
structure/provenance
only; weak-layer and
snowpack validation
remain next-phase work.

DECISION CLOSE

- The platform already provides a serious decision-support shell and disciplined governance posture.
- The missing science is visible rather than hidden, which makes the next research step concrete.
- **Meeting question:** should the scientist team and SASE/DGRE partner contacts help own the benchmark, validation, and promotion frontier from here?